

Realization of entire Burgers profiles by spatial patching

Matthew J. Colbrook

Department of Applied Mathematics and Theoretical Physics
University of Cambridge, Cambridge CB3 0WA, United Kingdom
m.colbrook@damtp.cam.ac.uk

2 October 2026

Abstract

We construct bounded initial data for the viscous Burgers equation whose translated late-time limits contain any prescribed profile represented by a compactly supported probability measure. A uniform fixed-time locality estimate for the Cole–Hopf formula allows backward histories to be placed in disjoint spatial windows. The construction also produces a single initial datum realizing every probability-measure profile in a fixed value interval.

Submission. AIM 351 at repository revision `fa98b7525fa3`; solution claim awaiting pull-request review.

1 Statement and result

Fix $\alpha < \beta$ and a Borel probability measure μ supported in $[\alpha, \beta]$. Write

$$\phi_\mu(x) = \frac{\int z e^{-zx/2} d\mu(z)}{\int e^{-zx/2} d\mu(z)}. \quad (1.1)$$

The target permits arbitrary observation shifts x_k and imposes no prescribed tails, monotonicity or spatial recurrence on the initial datum [1, 2]. That freedom is decisive.

Theorem 1.1. *There exists $u_0 \in L^\infty(\mathbb{R})$, with $\alpha \leq u_0 \leq \beta$ almost everywhere, such that the solution of*

$$u_t + uu_x = u_{xx}, \quad u(0, \cdot) = u_0,$$

has a sequence of observation points x_k satisfying

$$\sup_{|x| \leq k} |u(k, x_k + x) - \phi_\mu(x)| < \frac{1}{k}, \quad k \geq 1.$$

Consequently $u(k, x_k + \cdot) \rightarrow \phi_\mu$ uniformly on every compact interval.

2 A bounded entire history

Define, for every real s and x ,

$$H(s, x) = \int e^{-zx/2 + z^2 s/4} d\mu(z), \quad U(s, x) = -2 \frac{H_x(s, x)}{H(s, x)}. \quad (2.1)$$

Compact support permits differentiation to every order on compact (s, x) -sets. Moreover, $H > 0$ and $H_s = H_{xx}$. A direct differentiation of the logarithmic derivative gives

$$U_s + UU_x = U_{xx}.$$

The same formula writes $U(s, x)$ as a positive weighted average of $z \in [\alpha, \beta]$, hence

$$\alpha \leq U(s, x) \leq \beta \quad (s, x \in \mathbb{R}), \quad U(0, x) = \phi_\mu(x). \quad (2.2)$$

No finiteness or discreteness assumption on the support of μ was used.

3 A uniform fixed-time locality lemma

For $g \in L^\infty(\mathbb{R})$ set

$$h_g(y) = \exp\left(-\frac{1}{2} \int_0^y g(z) dz\right), \quad W_g(t, x) = \int_{\mathbb{R}} G_t(x - y) h_g(y) dy,$$

where $G_t(z) = (4\pi t)^{-1/2} e^{-z^2/(4t)}$. Define

$$S_t g(x) = -2 \frac{\partial_x W_g(t, x)}{W_g(t, x)}. \quad (3.1)$$

The Gaussian dominates the at-most-exponential growth of h_g , so these expressions and their positive-time derivatives exist.

For completeness, $h'_g = -gh_g/2$ almost everywhere. Integration by parts, whose boundary term vanishes by Gaussian decay, gives

$$S_t g(x) = \frac{\int G_t(x - y) g(y) h_g(y) dy}{\int G_t(x - y) h_g(y) dy}. \quad (3.2)$$

Thus $\alpha \leq S_t g \leq \beta$ whenever $\alpha \leq g \leq \beta$. The heat equation and logarithmic differentiation give the Burgers equation. The approximate-identity property gives the initial trace at Lebesgue points and in L^1_{loc} . This is the bounded Burgers solution; bounded well-posedness and the Cole–Hopf representation are recalled in [2, 3, 4].

Lemma 3.1 (Uniform locality). *For every $M, t, L, \varepsilon > 0$, there exists $R > 0$ such that, whenever*

$$\|g\|_\infty, \|h\|_\infty \leq M, \quad g = h \text{ almost everywhere on } [-R, R],$$

one has

$$\sup_{|x| \leq L} |S_t g(x) - S_t h(x)| < \varepsilon.$$

The radius depends only on M, t, L, ε , not on the exterior values of either datum.

Proof. The normalization at zero is important. It gives

$$e^{-M|y|/2} \leq h_g(y), h_h(y) \leq e^{M|y|/2}, \quad h_g = h_h \quad \text{on } [-R, R].$$

For $j = 0, 1$,

$$|\partial_x^j (W_g - W_h)(t, x)| \leq 2 \int_{|y| > R} |\partial_x^j G_t(x - y)| e^{M|y|/2} dy. \quad (3.3)$$

For fixed t, L, M , the right side tends to zero uniformly for $|x| \leq L$. Indeed, the Gaussian, multiplied by the polynomial arising from its first derivative and by $e^{M|y|/2}$, has an integrable majorant uniform on that compact x -interval.

There is also a uniform positive lower bound on both denominators. Restricting the heat integral to $-1 \leq y \leq 1$ yields

$$W_g(t, x), W_h(t, x) \geq m := \frac{2}{\sqrt{4\pi t}} \exp\left(-\frac{(L+1)^2}{4t} - \frac{M}{2}\right) > 0. \quad (3.4)$$

Also

$$|\partial_x W_h(t, x)| \leq B := \sup_{|x| \leq L} \int |\partial_x G_t(x - y)| e^{M|y|/2} dy < \infty.$$

Therefore

$$|S_t g - S_t h| \leq \frac{2|\partial_x(W_g - W_h)|}{m} + \frac{2B|W_g - W_h|}{m^2}.$$

Choosing R so that both Gaussian tails in (3.3) are sufficiently small proves the lemma. $\square$

4 Patching the backward histories into one datum

Proof of Theorem 1.1. Put $M = \max(|\alpha|, |\beta|) > 0$ and $g_k(x) = U(-k, x)$. By (2.2), all g_k take values in $[\alpha, \beta]$.

The semigroup identity needed here can be checked directly, without an appeal to uniqueness. The normalized Cole–Hopf datum of g_k is

$$h_{g_k}(x) = \frac{H(-k, x)}{H(-k, 0)}.$$

For $z \in [\alpha, \beta]$,

$$G_k * e^{-z(\cdot)/2}(x) = e^{-zx/2 + z^2 k/4}.$$

Tonelli’s theorem in (2.1) therefore gives

$$G_k * h_{g_k}(x) = \frac{H(0, x)}{H(-k, 0)}, \quad S_k g_k = \phi_\mu.$$

Use Lemma 3.1 with $t = k$, $L = k$ and $\varepsilon = 1/k$ to obtain a finite radius R_k . Choose centers x_k so that the intervals

$$I_k = [x_k - R_k, x_k + R_k]$$

are pairwise disjoint; for example, choose them recursively from left to right with gaps of length one. Fix any $c \in [\alpha, \beta]$ and define the single measurable function

$$u_0(y) = \begin{cases} g_k(y - x_k), & y \in I_k, \\ c, & y \notin \bigcup_{k \geq 1} I_k. \end{cases} \quad (4.1)$$

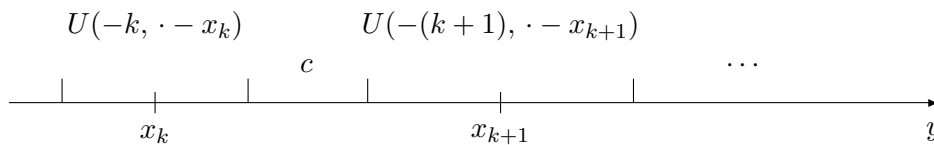


Figure 1: Disjoint initial-data windows on the real line, with constant datum c between them. Each radius is selected after its observation time. The drawing is schematic and is not to scale.

It satisfies the required pointwise bounds. For each k , the translated datum $y \mapsto u_0(x_k + y)$ agrees with g_k on $[-R_k, R_k]$. Translation invariance and the locality lemma yield

$$\sup_{|x| \leq k} |S_k u_0(x_k + x) - S_k g_k(x)| < 1/k.$$

Since $S_k g_k = \phi_\mu$, this is exactly the claimed estimate. $\square$

Remark 4.1 (The order of the choices). The radius R_k is allowed to be very large. It is chosen after the observation time k , not before it. Only then is the k th spatial window placed. There is no claim that heat propagation has finite speed, and there is no need to exchange a large-time limit with a large-radius limit. Uniform control of all possible exterior patches is precisely the content of Lemma 3.1.

5 A universal initial datum

Corollary 5.1 (Universal realization). *For fixed $\alpha < \beta$, one bounded initial datum can be chosen so that every ϕ_μ , with μ a probability measure on $[\alpha, \beta]$, occurs as a translated late-time limit.*

Proof. Choose a sequence (μ_k) in the compact metrizable space of probability measures on $[\alpha, \beta]$ such that every measure is a subsequential limit along indices tending to infinity. For example, enumerate a countable dense subset in successive repeated finite blocks. At stage k , patch the backward history $U_{\mu_k}(-k, \cdot)$, using the same tolerances as above. Then

$$\sup_{|x| \leq k} |u(k, x_k + x) - \phi_{\mu_k}(x)| < 1/k.$$

Weak convergence of measures implies convergence of the numerator and denominator in (1.1), uniformly on compact x -intervals. This follows from compactness of the z -support, uniform equicontinuity in x , and a uniform positive lower bound for the denominator. Passing to an appropriate subsequence proves the assertion. $\square$

References

- [1] M. Colbrook, *AIM 351: Can every bounded entire Burgers profile recur at late times?*, repository revision [fa98b7525fa3](#), consulted 2 October 2026. [Pinned problem statement](#).
- [2] T. Gallay and A. Scheel, *Viscous shocks and long-time behavior of scalar conservation laws*, Communications on Pure and Applied Analysis 23(10) (2024), 1448–1482. [arXiv:2306.13341v1](#). doi:10.3934/cpaa.2023119. See the conjecture following Proposition 6.2.
- [3] E. Hopf, *The partial differential equation $u_t + uu_x = \mu u_{xx}$* , Communications on Pure and Applied Mathematics 3 (1950), 201–230. doi:10.1002/cpa.3160030302.
- [4] J. D. Cole, *On a quasi-linear parabolic equation occurring in aerodynamics*, Quarterly of Applied Mathematics 9(3) (1951), 225–236. doi:10.1090/qam/42889.